Volume II · Chapter 11 — First Fine-Tunes (PubMedBERT, QLoRA, local MedGemma/Ollama) · Theory

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-II-Chapter-11-context.md` for the AI interaction log.

Prerequisites: Ch 7–10.

Learning outcomes — the student can: fine-tune a biomedical BERT for classification; apply QLoRA to fine-tune a small generative LLM on medical QA; run models locally via Ollama; evaluate a fine-tune and write a model card.

Topics (theory) — 10 h:

#	Topic	h
11.1	Fine-tuning mechanics: heads, learning rate, epochs, overfitting	2
11.2	Parameter-efficient fine-tuning: LoRA & QLoRA (quantization)	3
11.3	Datasets for fine-tuning: formatting, splits, tokenization	1
11.4	Local deployment: Ollama, GGUF, quantization levels	2

#	Topic	h
11.5	Evaluating a fine-tuned medical model; avoiding leakage	1
11.6	Compute & memory budgeting on consumer GPUs	1

Datasets/tools: HF `transformers`/`peft`/`bitsandbytes`, QLoRA, Ollama, MedGemma; GPU workstation. **Assessment:** fine-tuned model + model card (**60%**); eval report (**20%**); quiz (**20%**). **Key decisions:** full vs. PEFT fine-tuning; quantization level vs. quality; base model/size vs. hardware. **References:** plan §13 → *Fine-tuning…*; *Medical LLMs*; *…local inference*. **Hours:** Theory **10** + Lab **20** = **30**.